ID	Category	Description	Likelihood	Impact	Severity	Mitigation Strategy	Contigency Plan	NFR Reference
R-01	Technical	Booking logic produces incorrect results (e.g. overlapping check-in/check-out dates) due to missing validation	Medium	High	High	- Add input validation in BookingService for date ranges before any write	Revert to last known-good .java file via Git fix validation method re-run all tests before resubmission	NFR-053, NFR-054, NFR-055
R-02	Technical	Flat-file data corruption or data loss if the program crashes mid-write (e.g. partial booking record written to file)	Medium	High	High	- Write complete records atomically build the full record string in memory first, then write in a single operation - implement a backup copy of each data file before every write (NFR-058 equivalent)	- Keep a timestamped backup file (e.g. bookings_backup.txt) restore manually if the main file is corrupted - log all write operations	NFR-011, NFR-028
R-03	Technical	File I/O parsing errors crash the application when reading malformed or empty flat files (e.g. missing fields, wrong delimiters)	Medium	High	High	- Wrap all file reads in try-catch blocks - validate each parsed line before use - use a consistent delimiter (e.g. '|') and document the file format in the technical manual	- Skip and log malformed lines rather than crashing - Provide a file-repair utility or manual correction instructions in the user manual	NFR-011, NFR-013, NFR-028
R-04	Technical	Incompatibility across OS platforms (Windows/macOS/Linux) due to hardcoded file paths or OS-specific line endings (CRLF vs LF)	Medium	Medium	Medium	- Use File.separator and System.lineSeparator() throughout - Use relative paths only (NFR-033) - Test on at least two OS environments before submission	– Provide a path-configuration constant at the top of the main class so it can be changed in one place	NFR-030, NFR-031, NFR-033, NFR-038
R-05	Technical	Slow CLI response or poor UX due to inefficient flat-file reads (e.g. scanning entire bookings file on every search)	Low	Medium	Low	- Load data into in-memory collections (ArrayList/HashMap) at startup - operate on the in-memory model - Write back to file only on changes (NFR-003, NFR-004)	- Limit test data volume in demonstration - document known performance boundary (e.g. 'tested with up to 500 bookings') in the report	NFR-001, NFR-003, NFR-034, NFR-035, NFR-036, NFR-037
R-06	Security	Unauthorised access to owner/receptionist features due to missing or bypassable role checks in the CLI menu flow	Medium	High	High	- Never trust user-supplied role input directly - Log all login attempts to a text log file	Immediately push a patch to the affected menu class - Re-test all role paths - document the fix in the Git commit message	NFR-009, NFR-010
R-07	Security	Passwords stored in plain text in the flat user file, exposing credentials if the file is accessed directly	High	High	High	- Hash passwords using SHA-256 before writing to file - Never store or display plain-text passwords (NFR-007)	If hashing is not yet implemented, restrict file permissions on the users file and document this as a known limitation with a planned fix in the report	NFR-006, NFR-007
R-08	Project	Team member unavailability (illness, academic pressure) causing delays in Java code or report milestones	Medium	High	High	Maintain shared Git repository with frequent commits (NFR-026)	– Redistribute tasks Immediately - Communicate to professor early - Prioritise MVP features (booking, login, room listing) over advanced features	NFR-023, NFR-026
R-09	Project	Scope creep, adding CLI features beyond agreed requirements, delaying core deliverables	Medium	Medium	Medium	Freeze functional requirements after PM#2; any new feature must be approved by the full team and logged as a deviation in the Team Reflections appendix	-- Defer unimplemented features to the 'Future Enhancements' section of the report - Focus on agreed scope only	—
R-10	Project	Feedback from progress deliverables not integrated into final Java code or report, risking mark deduction	Low	High	Medium	Log all professor feedback after each PM meeting in Teams Channel and assign integration tasks with a completion date immediately after each review	- Conduct a final feedback audit one week before the 22 Apr deadline - allocate a buffer day specifically for integration of outstanding feedback	—
R-11	Usability	Poor CLI usability such as unclear menus, missing input validation feedback, or no confirmation dialogs, reducing UX marks in evaluation	Medium	Medium	Medium	- Display numbered menus with clear labels - Show inline error messages for all invalid inputs (NFR-013) - Add 'Are you sure? (y/n)' prompts before destructive actions like cancel/delete (NFR-020)	– Conduct a team walkthrough of all CLI flows before submission - fix any confusing menu paths - document the CLI navigation in the user manual	NFR-012, NFR-013, NFR-015, NFR-016, NFR-019, NFR-020
R-12	Compliance	No mechanism for users to access, correct, or delete their personal data stored in flat files, violating GDPR-aligned NFRs	Low	Medium	Low	- Implement 'View my data', 'Update my profile', and 'Delete my account' menu options for Guest users - deletion must remove or anonymise the user's record in the flat file (NFR-045/046/047)	– Add a manual admin-deletion procedure in the technical manual as a fallback - Document compliance approach in the report	NFR-045, NFR-046, NFR-047